

YINGDAN SHI

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EDUCATION

Illinois Institute of Technology

Ph.D. in Computer Engineering. Advisor: [Prof. Ren Wang](#)

Aug. 2024 - Present

GPA: 4.0/4.0

Soochow University

M.E. in Computer Technology. Advisor: Prof. Jingya Zhou

Sep. 2021 - Jun. 2024

GPA: 3.7/4.0

Hangzhou Normal University

B.E. in Computer Science and Technology.

Sep. 2017 - Jun. 2021

GPA: 3.8/4.0

RESEARCH INTERESTS

- Post-training, Agentic System, LLMs, MLLMs
- Agent Safety, Safety Alignment, Machine Unlearning, Continual Learning, Model Editing

PUBLICATIONS

(* co-first author)

- [1] [Yingdan Shi](#), Xiang Xu, Kaize Ding, Alfred O. Hero, Ren Wang. “On the Plasticity Collapse in Continual Machine Unlearning.” **ECCV**, 2026. [[Paper](#)][[Code](#)]
- [2] [Yingdan Shi](#), Ren Wang. “Exploring Nonlinear Pathway in Parameter Space for Machine Unlearning.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [3] [Yingdan Shi](#), Sijia Liu, Kaize Ding, Ren Wang. “Tackling Fake Forgetting through Uncertainty Quantification.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [4] [Yingdan Shi](#), Ren Wang. “Trajectory Unlearning on LLM-based Agents.” In submission, 2026. [[Paper](#)] [[Code](#)]
- [5] Jane Downer*, [Yingdan Shi*](#), Ziyang Liu*, Ren Wang, Binghui Wang. “Watermarking Graph Neural Networks via Explanations for Ownership Protection.” **ICML**, 2026. [[Paper](#)] [[Code](#)]
- [6] Duan Tu, [Yingdan Shi](#), Xu Zhang, Hongkang Li, Ren Wang. “Optimizing over Mode Connectivity for Multi-Task Learning and Machine Unlearning.” In submission, 2026.
- [7] Haoxin Guo, [Yingdan Shi](#), Ren Wang. “Beyond Superficial Metrics: Progressive Layer-wise Expert Unlearning for Deep Erasure in LLMs.” In submission, 2026.
- [8] [Yingdan Shi](#), Jingya Zhou, Congcong Zhang. “Towards Node-Level Influence Estimation in Social Networks.” In submission, 2025.
- [9] [Yingdan Shi](#), Jingya Zhou, Congcong Zhang, Zhenyu Hu. “Influence contribution ratio estimation in social networks.” *Information Sciences [J]*, 2025. [[Paper](#)]
- [10] [Yingdan Shi](#), Jingya Zhou, Congcong Zhang. “DySuse: Susceptibility Estimation in Dynamic Social Networks.” *Expert Systems with Applications [J]*, 2023. [[Paper](#)]
- [11] Zhaoying Wang, [Yingdan Shi](#), Xiang Liu, Can Chen, Jun Wen, Ren Wang. “HODDI: A Dataset of High-Order Drug-Drug Interactions for Computational Pharmacovigilance.” In submission, 2025.
- [12] Zhenyu Hu, Jingya Zhou, Wenqi Wei, Congcong Zhang, [Yingdan Shi](#). “Predicting Cross-domain Collaboration Using Multi-task Learning.” *Expert Systems with Applications [J]*, 2024. [[Paper](#)]
- [13] Zhenyu Hu, Jingya Zhou, Congcong Zhang, [Yingdan Shi](#). “Explainable Cross-Domain Collaborator Recommendation.” *CSCWD 2024*.
- [14] Congcong Zhang, Jingya Zhou, Jin Wang, Jianxi Fan, [Yingdan Shi](#). “Fairness-Aware Competitive Bidding Influence Maximization in Social Networks.” *IEEE Transactions on Computational Social Systems [J]*, 2024.
- [15] Congcong Zhang, Jingya Zhou, Wenqi Wei, [Yingdan Shi](#). “Order-sensitive Competitive Revenue Maximization for Viral Marketing in Social Networks”. *Information Sciences [J]*, 2025.

RESEARCH EXPERIENCE

TIML, Illinois Institute of Technology

Research Assistant. Advisor: [Prof. Ren Wang](#)

Chicago, IL

Aug. 2024 - Present

- **Self-unlearning on Self-evolving Agents.** We identified a critical vulnerability in self-evolving agents: forgotten knowledge can be inadvertently recovered during self-evolution. We propose integrating self-unlearning into the self-evolution process to prevent the memorization of forgotten information.
- **Trajecotry Unlearning on LLM-based Agents.** Investigated the gap between agent unlearning and conventional LLM unlearning, demonstrating that existing LLM unlearning methods are insufficient for achieving safe and effective unlearning in agent systems and preventing undesirable behavior.
- **Plasticity Collapse in Continual Unlearning.** Identified *Unlearning Plasticity Collapse*, a novel phenomenon in continual unlearning with two failure modes: *forward failure*, where the model progressively loses its ability to forget new data, and *backward failure*, where previously forgotten data is re-memorized. Theoretical analysis attributes this to accumulating geometric constraints in parameter space, with empirical validation across image classification, image generation (DDPM), and multimodal tasks (LLaVA).
- **LLM Knowledge Entanglement.** Rather than representing knowledge as a single vector, proposed characterizing knowledge via the subspace of hidden states and systematically investigating inter-knowledge entanglement. Leveraging this entanglement structure provides principled guidance for knowledge unlearning in LLMs.
- **Mode Connectivity Unlearning.** Revealed that existing linear unlearning methods suffer from a weight entanglement issue that prevents ideal forgetting. Exploited mode connectivity to explore an unlearning pathway from a nonlinear perspective, enabling plug-and-play integration with existing unlearning methods and mitigating both over-forgetting and under-forgetting.
- **Fake Forgetting.** Observed that existing machine unlearning methods overlook prediction uncertainty. Through conformal prediction, we show that data points diagnosed as forgotten based on unlearning accuracy can still be exposed, revealing a critical gap in current evaluation. Proposed a novel uncertainty-aware metric and framework to address this limitation.
- **GNN Watermarking.** Identified vulnerabilities in backdoor-based watermarking, including ownership ambiguity and susceptibility to data poisoning. Proposed an explanation-based scheme that eliminates data manipulation and dependency while preserving the benefits of backdoor-based methods.

WORK EXPERIENCE

Illinois Institute of Technology

Research Assistant

Chicago, IL

Aug. 2024 - Present

Meituan Inc.

Business Analyst Internship

Beijing, China

May. 2024 - Jul. 2024

Soochow University

Research Assistant

Suzhou, China

Aug. 2021 - Jun. 2024

AWARDS AND HONORS

Excellent Master's Thesis Award (2024)

Outstanding Master's Award (2024)

National Scholarship (2023)

Outstanding Undergraduate Award (2021)

First Prize Scholarship (2020,2021,2022)

ACADEMIC SERVICE

Conference Reviewer: ICML 2026, SIGIR 2026, NeurIPS 2026

Journal Reviewer: IEEE Transactions on Dependable and Secure Computing, World Wide Web, Scientific Reports, Dynamic Games and Applications, Humanities and Social Sciences Communications, Knowledge and Information Systems, Social Network Analysis and Mining